STACKS, DISCRIMINANTS: LEMMA-EXPLAIN-CONDITION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3187-3206 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 3208-3243, the complete author proof (34 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_discriminant.json (pb=1731, stmt 3187-3206, proof 3208-3243).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00LD: sources/stacks-discriminant/pinned-original-source.tex, source lines 3243–3243. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0312: sources/stacks-discriminant/pinned-original-source.tex, source lines 3243–3243. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00HT: sources/stacks-discriminant/pinned-original-source.tex, source lines 3243–3243. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a map of Noetherian rings. Consider the conditions*

(1) *nonzerodivisors of A map to nonzerodivisors of B,*
(2) *(1) holds and $Q(A) \to Q(A) \otimes_A B$ is flat,*
(3) *$A \to B_{\mathfrak{q}}$ is flat for every $\mathfrak{q} \in Ass(B)$,*
(4) *(3) holds and $A \to B_{\mathfrak{q}}$ is flat for every $\mathfrak{q}$ lying over an element in $Ass(A)$.*

Then we have the following implications

$$\begin{array}{ccc} (1) & \Longleftarrow & (2) \\ \Uparrow & & \Downarrow \\ (3) & \Longleftarrow & (4) \end{array}$$

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

If going up holds for $A \to B$ then (2) and (4) are equivalent.

Proof. The horizontal implications in the diagram are trivial. Let $S \subset A$ be the set of nonzerodivisors so that $Q(A) = S^{-1}A$ and $Q(A) \otimes_A B = S^{-1}B$. Recall that $S = A \setminus \bigcup_{\mathfrak{p} \in \mathrm{Ass}(A)} \mathfrak{p}$ by Algebra, Lemma 00LD. Let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{p} \subset A$.

Assume (2). If $\mathfrak{q} \in \mathrm{Ass}(B)$ then $\mathfrak{q}$ consists of zerodivisors, hence (1) implies the same is true for $\mathfrak{p}$. Hence $\mathfrak{p}$ corresponds to a prime of $S^{-1}A$. Hence $A \to B_\mathfrak{q}$ is flat by our assumption (2). If $\mathfrak{q}$ lies over an associated prime $\mathfrak{p}$ of A, then certainly $\mathfrak{p} \in \mathrm{Spec}(S^{-1}A)$ and the same argument works.

Assume (3). Let $f \in A$ be a nonzerodivisor. If f were a zerodivisor on B, then f is contained in an associated prime $\mathfrak{q}$ of B. Since $A \to B_\mathfrak{q}$ is flat by assumption, we conclude that $\mathfrak{p}$ is an associated prime of A by Algebra, Lemma 0312. This would imply that f is a zerodivisor on A, a contradiction.

Assume (4) and going up for $A \to B$. We already know (1) holds. If $\mathfrak{q}$ corresponds to a prime of $S^{-1}B$ then $\mathfrak{p}$ is contained in an associated prime $\mathfrak{p}'$ of A. By going up there exists a prime $\mathfrak{q}'$ containing $\mathfrak{q}$ and lying over $\mathfrak{p}$. Then $A \to B_{\mathfrak{q}'}$ is flat by (4). Hence $A \to B_\mathfrak{q}$ is flat as a localization. Thus $A \to S^{-1}B$ is flat and so is $S^{-1}A \to S^{-1}B$, see Algebra, Lemma 00HT. □